

Dear Editor,

Thank you for giving us the opportunity to revise and resubmit our protocol paper:

Effects of Advanced Trauma Life Support® Training Compared with Standard Care on Adult Trauma Patient Outcomes (ADVANCE TRAUMA): Study Protocol for a Stepped-Wedge Cluster Randomised Trial.

Our point-by-point response is below. We believe that addressing these comments significantly improved our paper and we thank the reviewers for taking time to read and comment on our paper.

On behalf of all authors,

Martin Gerdin Wärnberg

Reviewer reports

Reviewer #1

Q1.1 Thank you for addressing the points highlighted in the previous review.

R1.1 Thank you for reviewing our manuscript!

Reviewer #2

Peer review of: Effects of Advanced Trauma Life Support® Training Compared with Standard Care on Adult Trauma Patient Outcomes (ADVANCE TRAUMA): Study Protocol for a Stepped-Wedge Cluster Randomised Trial

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Manuscript ID: TRLS-D-25-01007_R1^[1] Overall recommendation: major revision

Summary

Q2.1 This is a well-motivated, ambitious batched stepped-wedge cluster randomised trial of ATLS® training versus standard care in 30 hospitals across India, with the primary outcome of in-hospital mortality within 30 days. The protocol is clear in many parts (design, outcomes, oversight) and addresses an important knowledge gap because high-quality evidence for ATLS®'s effect on patient outcomes is lacking.

Major strengths

1. Important clinical question and large scale. The trial addresses a major evidence gap about ATLS® and trauma outcomes; the planned 30 clusters and 4,320 participants provide potential for informative, policy-relevant results.

2. Appropriate and thoughtful design. Use of a batched stepped-wedge and nested staircase for intensive secondary outcome measurement is pragmatic and justified given logistics and training constraints.

3. Comprehensive oversight and intention for open data. The trial has Trial Management and Steering/DMC structures and plans to make protocol, code, and deidentified data available.

R2.1 Thank you for a thorough and helpful review of our manuscript. Your comments really helped us improving the manuscript and we are hopeful that this trial, once complete, will provide meaningful evidence to strengthen the care of trauma patients globally.

Major concerns — please address before submission

1) Primary outcome definition and ascertainment — practical risks and bias

Issue: The primary outcome is "in-hospital mortality within 30 days of arrival". For patients transferred between hospitals the protocol relies on telephone follow-up or contacting the receiving hospital. This risks differential ascertainment (e.g., loss to follow-up differing by cluster/time) and potential misclassification.

Suggested actions (required):

Q2.2 Provide concrete procedures and responsibilities for ascertaining 30-day status for transferred/discharged patients (who calls, how many attempts, what records are checked). State acceptable sources of confirmation (hospital EHR, death registry, phone with next-of-kin) and planned documentation.

R2.2 We have now clarified that follow-up of patient participants is conducted by the Clinical Research Coordinator (CRC) at each site. The site coordinators regularly review clinical notes and hospital registers to determine whether participants have been discharged or transferred. For participants who have been transferred, the CRC conducts telephonic follow-up calls with either the participant or their next of kin.

Deaths are confirmed through verification of hospital records. For outcome assessments such as EQ-5D, WHODAS, and return-to-work status, the CRC makes regular follow-up calls to obtain updates from the participant or their nearest kin. By Day 30, at least three to four phone calls are attempted. Depending on the response, a subsequent telephone follow-up will be scheduled. In cases where no contact can be established despite repeated attempts, the participant is classified as a loss to follow-up.

Q2.3 Prespecify how you will handle deaths that cannot be confirmed at 30 days (e.g., treat as alive, treat as missing — and how this choice will be handled in sensitivity analyses). Add an analysis plan for differential loss to follow-up (e.g., inverse-probability weighting or worst-case/best-case sensitivity analyses).

R2.3 In-hospital mortality is unlikely to be missing. Patients who are discharged alive before day 30 will be considered alive. For all-cause mortality at 30 days and beyond, deaths that cannot be confirmed will be treated as missing. We will handle missing data in the primary outcome depending on the amount of missing. If there is less than 10% missingness in the primary outcome, then we will conduct a complete case analysis. If there is more than 10% missingness then we will use multiple imputation, unless we have reason to believe that the

data may be missing not at random. We have clarified this in the manuscript, item 12 and 20c.

2) Blinding and outcome assessment bias

Issue: No blinding of outcome assessors or data analysts is planned because of time-randomised intervention. While understandable, this raises the possibility of ascertainment bias for outcomes that require judgement (e.g., ICU admission criteria, adherence assessment).

Suggested actions:

Q2.4 For objective outcomes (mortality) ensure independent verification where feasible (e.g., central adjudication of ambiguous deaths).

R2.4 In case death occurs in the hospital, a hundred percent review of the source data is done through onsite and remote monitoring. This has been clarified in the manuscript, item 18a.

Q2.5 For outcomes subject to ascertainment (ICU admission, length of stay), describe standardised, objective definitions and whether any central adjudication will occur.

R2.5 See R2.4. In addition, ICU admission and length of stay are defined and available from the medical records.

Q2.6 If possible, blind the statistician performing the primary analysis to intervention timing (e.g., anonymised period labels) or document why blinding the analyst is infeasible and how you will mitigate bias.

R2.6 For stepped wedge trials it is difficult to blind the analysing statistician since treatment and time are related to each other. Instead, to protect against issues to do with an unblinded analysis of the data, a full statistical analysis plan – including example code and descriptions of all sensitivity analyses – will be published online prior to the statistician receiving any data. This has been clarified in item 17a.

3) Statistical analysis — small number of clusters and choice of corrections

Issue: The protocol recognises the small cluster number (30) and plans a "correction for a small number of clusters" but does not pre-specify which method (e.g., Kenward-Roger, permutation, cluster-robust variance with small-sample correction, bootstrap).

Suggested actions (required):

Q2.7 Pre-specify the primary small-sample correction method (or provide a decision algorithm with ranked preferred methods and justification). Common choices: (a) use cluster-level summaries with permutation tests, (b) use GLMM with Kenward-Roger or Satterthwaite df, or (c) use bias-corrected cluster-robust standard errors with a recommended correction (e.g., Webb/Imbens). Cite supporting references.

R2.7 What small-sample correction method to use is an active area of research, particularly in the context of stepped-wedge and batched stepped-wedge designs. We have now specified

that we will use a Kenward Roger correction as our primary method, but will potentially update this choice if there is new methods work supporting other methods closer to the end of patient inclusion. This has been clarified under item 20a.

Q2.8 Clearly state the primary estimand (risk difference or risk ratio/OR) and primary modelling choice (e.g., binomial identity model for risk difference; if non-convergence, report odds ratio). You already mention this — but make it explicit which effect will be the main inference and how small-cluster bias will be corrected.

R2.9 The primary estimand will be the odds ratio. This has been clarified under item 20a.

4) Handling of cluster size imbalance and time trends

Issue: Stepped-wedge designs are sensitive to secular trends and cluster size imbalance. While the protocol will adjust for period effects, more detail is needed on preplanned checks and adjustments.

Suggested actions:

Q2.9 Prosperity diagnostics for secular trends (plots of outcome rates by period), and the approach if strong secular trends are present (e.g., spline for time, additional covariate adjustment). You mention the spline option — indicate thresholds that would trigger spline modelling.

R2.9 Period effects will be adjusted for in all models and this will account for secular trends. We will however conduct the analysis with the time effect modelled using a spline function regardless of whether we think that a strong secular trend is present, thus there is no need to specify thresholds.

Q2.10 Describe how you will account for variable cluster sizes in sample-size re-estimation or analysis (you mention allowing for variation — state if any cluster-weighted analyses or robust SEs will be used).

R2.10 We will not re-estimate the sample size in the interim analysis, see section 21b. Variable cluster sizes were accounted for in the original sample size calculation, based on data from previous projects. Also, our chosen analysis approach (generalised linear mixed models) does not require any explicit consideration of how varying cluster sizes will be accommodated at the analysis stage. Nonetheless, we might anticipate that the intervention effect might differ in smaller vs larger hospitals and so we have therefore added a sub-group analysis in which we will investigate treatment effect heterogeneity by cluster size,

5) Interim analysis stopping rules and multiplicity

Issue: An interim analysis is planned for feasibility after half the batches complete, but no stopping rules or decision criteria are specified (e.g., for harm, efficacy, futility).

Suggested actions:

Q2.11 Clarify the exact objectives of the interim review and prespecified quantitative criteria that would trigger stopping (for feasibility) and how the SDMC will make recommendations.

If no efficacy stopping is planned, state explicitly. If harm could prompt stopping, set specific thresholds or statistical boundaries (or explain why not applicable).

R2.11 The purpose of the interim analysis is to stop the trial if it is deemed not feasible. This will be a qualitative decision based on an overall assessment of the trial's progress so far. We now explicitly state that no efficacy stopping is planned and provide a justification, see item 21b.

6) Consent and ethical considerations

Issue: The protocol uses opt-out consent for routine data and opt-in for non-routine data; waiver for unconscious patients is also planned. While justified as minimal risk, opt-out and waiver approaches require detailed local justification and documentation.

Suggested actions:

Q2.12 Include the exact text (or a summary) of patient information sheets and how opt-out is communicated at arrival (timing, language, signage). State whether local ethics committees approved opt-out/waiver approaches (you indicate approvals but list only 1-2 bodies). Provide the ethical approval reference numbers in the main text.

R2.12 The English text is available below. Since the trial is pan India, the consent form has been translated into the regional languages. The CRC takes the consent in the local languages, which the participants or its relatives are in comfortable with. The local ethics committees have approved the consent model.

Text from information sheet:

You are invited to participate in a research study. Before deciding to take part, it's important to understand the purpose of this research and what it involves. Please read the following information carefully and feel free to discuss the study with others if you wish. If anything is unclear or if you would like more information, do not hesitate to ask us. Please take your time to decide whether you wish to participate.

Purpose of the Study and Potential Outcomes

1. What is the purpose of the study?

We are conducting a study at this hospital to determine whether special training, known as Advanced Trauma Life Support (ATLS), provided to doctors, improves the care and outcomes for trauma patients.

2. What does the intervention involve?

You will not receive any direct intervention as part of this study. The intervention consists of a two-and-a-half-day specialized training course for the doctors who will be treating you.

3. Why have I been invited?

You have been invited to participate in this study because you were admitted to this hospital after an injury. We plan to include more than 4,000 participants from 30 hospitals across India.

4. What will happen if I participate?

During your hospital stay, we will collect certain information required for this study from your routine medical records. This includes your age, gender, medical history, details of your injury, and the treatment you received. During the study's follow-up phase, the hospital research team will:

- Call you or a relative one day after your arrival at the hospital to check on your well-being. If you are still in the hospital, we may visit you in the ward.
- Call you or a relative 30 days after your arrival to check on your well-being. We may visit if you are still in the hospital.
- Call you or a relative three months (90 days) after your arrival to check on your well-being. If you are still in the hospital, we may visit you in the ward. During this time, we may also conduct a short interview, lasting less than five minutes, to inquire about your health status.

5. Do I have to participate?

Participation in this study is entirely voluntary. For the follow-up phase of this study (after hospital discharge), we may request you or your relative to sign an Informed Consent Form. However, you are free to change your mind and request us to stop collecting follow-up data at any time without providing a reason. Choosing not to be contacted for follow-up data collection will not affect your future medical treatment or standard care in any way. Any data already collected from your medical records will continue to be used in the analysis, unless you or your relative ask the site research team not to use the previously collected data by signing an 'opt-out' form, which is enclosed with this information sheet. The details of the site research team are provided in this document.

6. What will I be required to do?

You will be asked to provide responses to questions about your health and recovery during telephonic follow-ups or in-person follow-up visits.

7. What are the side effects of the intervention?

There are no foreseeable side effects of this intervention, as no direct intervention will be administered to you.

8. What are the potential disadvantages and risks of participating?

We have not identified any risks associated with participating in this study.

9. What are the possible benefits of participating?

Our research may help to discover tools that better identify injured patients in need of immediate care. While this research will not affect the care you receive in this hospital, its results may benefit others who are injured in the future. There is no guarantee that you will benefit directly from this study. However, your participation will contribute to medical knowledge about the use of this intervention.

10. Can I be compensated for participating?

You will not receive any payment for participating in this study.

11. What if I decide I no longer wish to participate?

You are free to withdraw from this study at any time without needing to provide a reason. Your decision will not affect your future care or medical treatment.

12. Will my participation in this study be kept confidential?

All information collected about you as part of your participation in this study will be kept strictly confidential and used solely for research purposes. Your information will be anonymized and stored on a highly secured server. All de-identified information about you will be managed in accordance with Indian data protection laws to ensure the utmost confidentiality. At the end of the study, your anonymized and grouped data may be shared with researchers outside the George Institute for Global Health, both in India and internationally, to further advance knowledge on injury care.

13. What will happen to the results of this study?

The findings of this research may be published in scientific journals; however, it will not be possible to identify you from any published article. Additionally, data from this project may be shared with other researchers in India and abroad without revealing your identity.

14. Who is funding this study?

This study is supported by the Karolinska Institutet, Sweden, through a grant from the Swedish Research Council.

If you have questions about this research study, you may contact the Principal Investigator.

Opt out summary: 1. I hereby request deletion of any routine medical information previously collected about me (or my relative) and ask that no further data be collected for this study.

2. I also request that we not be contacted — by phone or in person — for any follow-up data collection.

Q2.13 Clarify process for obtaining proxy consent and for post-recovery consent from previously unconscious participants, including time windows and how refusal would be handled for already collected data.

R2.13 Participant consent is obtained by the designated CRC at each site, either from the participant or from the nearest kin, depending on the participant's level of consciousness. If consent is obtained from the nearest kin, re-consent will be sought from the participant once they are in a condition to provide consent.

During the consent process, the CRC explains the study details and the data collection procedures that have been carried out since the participant's hospital admission. If the participant agrees to continue in the study, the participant or their nearest kin is informed about additional non-routine data collection procedures involved in the study.

If the participant or their nearest kin declines participation, they are given the option to remove any previously collected data or to stop further data collection (opt-out). Due to the nature of the trial, it is challenging to define a specific timeline for obtaining consent, as it largely depends on the availability of the participant or their relative, but it occurs before discharge.

7) Adherence (fidelity) measurement

Issue: Adherence to ATLS® principles will be measured using a 14-item checklist by observers. Observer measurement risks Hawthorne effects and inter-observer variability. Suggested actions:

Q2.14 Describe observer training, inter-rater reliability assessments (e.g., kappa targets) and whether observers will be independent and blinded to period where possible.

R2.14 Adherence is assessed by trained CRCs. The training involve online sessions on ATLS procedures using the designated checklist. Before collecting this data for the trial, CRCs collect pilot data from ten patients under the supervision of a local investigator or senior clinician. This pilot data is then discussed with the trial team. We do not formally assess inter-rater reliability for this secondary outcome. The CRCs are not blinded to period.

Q2.15 State plans to minimise Hawthorne bias (e.g., discreet observation, sampling across shifts) and how missing adherence observations will be handled.

R2.15 The busy, crowded and noisy environment where the observations are taking place help limit Hawthorne bias. The nested staircase design and sampled shifts also helps in limiting this bias. Missing adherence observations will be handled as missing, and managed using the methods described under item 20c.

8) Harms definition and reporting

Issue: Protocol excludes routine AE/SAE reporting because events are common in critically ill patients but identifies a small set of "safety events". The approach risks underreporting intervention-related harms or unintended consequences (e.g., delays in care due to added tasks).

Suggested actions:

Q2.16 Define a more comprehensive, preplanned list or criteria for safety signal detection beyond the three items listed, and specify how increases will be assessed (statistically vs. clinically). Clarify reporting timelines and responsibilities.

R2.16 The concept of safety events is not unique to this trial, see for example the cited STEPCARE trial protocol (citation 50). A more comprehensive list of potential complications would have been ideal, however our pilot study showed that collecting complication data with any reliability was not feasible. We therefore rely on these higher level safety signals and the investigators assessments of these as the basis for our safety monitoring. The CRCs or site investigators initiate the safety reporting and the site investigator is responsible for making a first assessment of whether the event is likely to be associated with the patient's condition or the trial procedure. If the event is assessed as associated with the trial, then the event is reviewed by the trial management group and then by the steering and data monitoring committee. Each instance has 24-48 hours to make this assessment. Regarding delays in care we are collecting emergency department length of stay as a secondary outcome and will thus be able to assess potential delays that result in a longer stay in the emergency department.

Minor issues and editorial suggestions

Q2.17 Abstract length & clarity: Shorten the "Methods" wording to be more compact (e.g., explicit numbers of clusters, batch structure, primary outcome). The abstract currently repeats some design details found in the body.

R2.17 We have shortened the methods section accordingly.

Q2.18 Trial dates and registration: Include trial registration numbers and registration dates in the abstract (you list CTRI and ClinicalTrials.gov in the document; ensure both IDs and dates are present in abstract and the journal's required format).

R2.18 We have ensured that both IDs and dates are mentioned.

Q2.19 Terminology consistency: Use consistent terms for time windows (e.g., "30-day in-hospital mortality" vs "in-hospital mortality within 30 days") across abstract, methods, and outcomes table.

R2.19 We have reviewed the manuscript for terminology consistency.

Q2.20 Tables and Figures: Ensure all tables and figures are included in the submission package (I note references to Figures 1-3 and Tables 1-3; verify high-resolution figure files and that figure captions are self-contained).

R2.20 We have ensured that tables and figures are included according to the journal's submission guidelines.

Q2.21 Competing interests: Several contributors are ATLS® instructors — consider describing mitigation (e.g., independent data analysis, sponsor non-involvement).

R2.21 The data analysis will be conducted by an independent trial statistician. This is an investigator initiated trial, and the principal investigator is not an ATLS instructor.

Q2.22 Grammar / small edits: A light copy edit is needed for typos and phrasing (e.g., "Thee are mainly" → "There are mainly"; consistent use of ATLS® trademark format).

R2.22 Thank you for pointing this out, we have submitted the manuscript for professional proof-reading before submission.